

Scenario 1: Employee Salary Analysis

Question:

Find the names and salaries of employees who earn more than the average salary in the company.

Explanation:

1. Calculate the average salary using aggregate function AVG()
2. Use WHERE clause to filter employees whose salary is more than average salary
3. Use SELECT to display the employee's name and salary

Scenario 2: Customer Orders without Matching Records

Question:

Retrieve a list of customer names who have not placed any orders.

Explanation:

1. Use LEFT JOIN between customers and orders table using customer_name.
2. Filter records using WHERE clause orders.customer_id is NULL to find without any matching orders.

Scenario 3: Product Sales Summary

Question:

Display the total sales amount for each product.

Explanation:

1. Use INNER JOIN between the products table and the sales table using product_id.
2. Use the SUM() function to calculate the total sales amount for each product.
3. Apply a GROUP BY clause on product_name to aggregate the sales data.
4. Display product_name and the total sales amount.

Scenario 4: Department-Wise Employee Count

Question:

List each department name with the number of employees working in it.

Explanation:

1. Perform LEFT JOIN between the employees table and the departments table using the common column department_id.
2. Use the COUNT() function to count the number of employees in each department.
3. Apply a GROUP BY on the department_name to aggregate the data.
4. Select the department name and the employee count.

Scenario 5: Top 3 Highest Sales

Question:

Find the top 3 highest sales transactions.

Explanation:

1. Select the necessary columns like sale_id, customer_id, and amount from the sales table.
2. Use the ORDER BY clause on the amount column in descending order to list the highest sales first.
3. Use LIMIT 3 to restrict the output to the top 3 records.

Scenario 6: Calculate Employee Salary Ranks by Department

You have a table employees with the following columns:

employee_id (Primary Key), name, age, salary, department_id

You also have a table departments with the following columns:

department_id (Primary Key), department_name

Task:

Write a query to display each employee's name, department name, salary, and their salary rank within their respective department.

Logic:

1. The rank should be calculated within each department.
2. Join (INNER JOIN) the employees table and departments table using the common column department_id to get the department name.
3. Use the **RANK()** window function to rank employees based on their salary within each department.
4. Display employee details with their salary rank using the **ORDER BY** clause.